

Math 237B Final

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Problem 1. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes such that $g \circ f$ is proper, g is quasi-projective and f is surjective. Prove or disprove that g is proper.

Solution. We'll prove that g is proper. We'll prove relevant statements in the following order:

- (a) g is separated, and f is proper.
- (b) g is of finite type.
- (c) g is universally closed.

Finally, use these results we can conclude that g is proper.

Proof of (a):

Since $g : Y \rightarrow Z$ is quasi-projective, there exists some scheme Y' , together with open immersion $\iota : Y \hookrightarrow Y'$ and projective morphism $\pi : Y' \rightarrow Z$, such that the following commutative diagram holds:

$$\begin{array}{ccc} Y & \xhookrightarrow{\iota} & Y' \\ g \downarrow & \swarrow \pi & \\ Z & & \end{array}$$

Which, recall that any open immersion is separated, and projective morphisms are proper (in particular separated), so $g = \pi \circ \iota$ is also separated.

As a consequence, since $g \circ f$ is proper, while g is separated, we concluded that f is proper also.

Proof of (b):

Let's show g is of finite type: Using the diagram in part (a), since $\pi : Y' \rightarrow Z$ is projective, it is proper (hence of finite type), so for any open affine subset $U \subseteq Z$ (say $U \cong \text{Spec}(A)$), such that $\pi^{-1}(U) = \bigcup_{i=1}^n V_i$, where each V_i is affine open subset, and $V_i \cong \text{Spec}(B_i)$ for some B_i that's finitely generated A -algebra.

Then, with open immersion $\iota : Y \hookrightarrow Y'$, one has $\iota^{-1}(\pi^{-1}(U)) = Y \cap \pi^{-1}(U) = \bigcup_{i=1}^n Y \cap V_i$. Since Y is open in Y' , each $Y \cap V_i$ is an open subset of V_i under its subspace topology, hence each $Y \cap V_i = \bigcup_{j \in J_i} D(f_{ij})$ for some $f_{ij} \in B_i$ (running over some index set J_i). Because each $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$ (which $(B_i)_{f_{ij}}$ is a finitely generated B_i -algebra), then with B_i being a finitely generated A -algebra, $(B_i)_{f_{ij}}$ is a finitely generated A -algebra also.

As a result, one has $g^{-1}(U) = \iota^{-1}(\pi^{-1}(U)) = \bigcup_{i=1}^n Y \cap V_i = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, where each $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$ has $(B_i)_{f_{ij}}$ being a finitely generated A -algebra.

Finally, since $g \circ f$ is finite type, it's quasi-compact, hence with affine open subset $U \subseteq Z$, one has $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$ being quasi-compact. However, since f is surjective, one has $g^{-1}(U) =$

$f(f^{-1}(g^{-1}(U)))$. Because it's an image of a quasi-compact set under continuous map, $g^{-1}(U)$ is quasi-compact. Hence, the previous cover for $g^{-1}(U)$, say $g^{-1}(U) = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, can be chosen as finite. This shows that g is of finite type.

Proof of (c):

To prove g is universally closed, for any morphism of schemes $h : Z' \rightarrow Z$ consider the following commutative diagram of multiple fibre products:

$$\begin{array}{ccc}
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 f' \downarrow & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 g' \downarrow & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

Which:

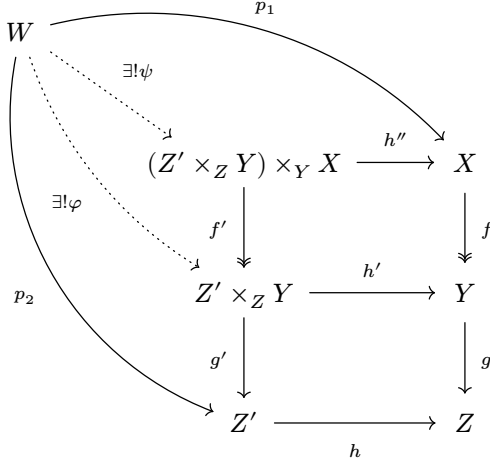
- $g' : Z' \times_Z Y \rightarrow Z'$ denotes the base change of $g : Y \rightarrow Z$ by $h : Z' \rightarrow Z$.
- $f' : (Z' \times_Z Y) \times_Y X \rightarrow Z' \times_Z Y$ denotes the base change of $f : X \rightarrow Y$ by $h' : Z' \times_Z Y \rightarrow Y$ (the pullback of $h : Z' \rightarrow Z$), which the surjectivity of f ensures the surjectivity of f' . And, since part (a) shows f is proper (which is universally closed), then f' is closed.

Now, we ought to show $(Z' \times_Z Y) \times_Y X$ together with $g' \circ f'$ and h'' satisfies the universal property of the fibre product $Z' \times_Z X$. For any scheme W together with two morphisms $p_1 : W \rightarrow X$ and $p_2 : W \rightarrow Z'$, such that $(g \circ f) \circ p_1 = h \circ p_2$. Then, the universal property of $Z' \times_Z Y$ guarantees a unique morphism of schemes $\varphi : W \rightarrow Z' \times_Z Y$, such that $p_2 = g' \circ \varphi$, and $f \circ p_1 = h' \circ \varphi$, as follow:

$$\begin{array}{ccc}
 W & \xrightarrow{p_1} & X \\
 \downarrow \varphi & & \downarrow f \\
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 f' \downarrow & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 g' \downarrow & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

(Note: In the diagram above, a curved arrow labeled $\exists! \varphi$ points from W to $Z' \times_Z Y$, and a curved arrow labeled p_2 points from W to Z' . The horizontal arrow from W to X is labeled p_1 .)

Furthermore, with the universal property of $(Z' \times_Z Y) \times_Y X$, the equality $f \circ p_1 = h' \circ \varphi$ guarantees a unique morphism of schemes $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$, such that $\varphi = f' \circ \psi$ and $p_1 = h'' \circ \psi$, as follow:



This proves both the existence and the uniqueness of the morphism $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$, such that $p_2 = g' \circ \varphi = (g' \circ f') \circ \psi$, and $p_1 = h'' \circ \psi$ (as any $\psi' : W \rightarrow (Z' \times_Z Y) \times_Y X$ with such property is enforced to be equal to ψ , based on individual fibre product's property). So, $(Z' \times_Z Y) \times_Y X$ indeed satisfies the universal property of $Z' \times_Z X$, showing that $g' \circ f'$ is the base change of the morphism $(g \circ f) : X \rightarrow Z$. Which, since $g \circ f$ is proper, $g' \circ f'$ is closed.

Finally, with f' and $g' \circ f'$ being closed, and f' being surjective, we claim that g' is closed:

For any closed subset $V \subseteq Z' \times_Z Y$, by the continuity of f' , the subset $(f')^{-1}(V) \subseteq (Z' \times_Z Y) \times_Y X$ is closed. Hence, $(g' \circ f')((f')^{-1}(V)) \subseteq Z'$ is closed (by closeness of $g' \circ f'$).

Also, since f' is surjective, one simply has $f'((f')^{-1}(V)) = V$. Hence, the above set equality shows that $(g' \circ f')((f')^{-1}(V)) = g'(V) \subseteq Z'$ is closed. This proves that g' is closed.

With g' being base change of g over arbitrary morphism of schemes $h : Z' \rightarrow Z$, this shows that g is universally closed.

With all information above, we can conclude that g is separated, finite type, and universally closed, hence proper.

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Problem 2. Let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be projective morphisms. Prove or disprove that the projection map $X \times_S Y \rightarrow S$ is projective.

Solution. We'll show that $X \times_S Y \rightarrow S$ is projective. We'll prove the following two statements:

- (a) Base change of projective morphisms are projective.
- (b) Composition of projective morphisms are projective.

Then, the two will help conclude the desired result.

Proof of (a):

First, we'll show that base change preserves projective spaces: More precisely, given any $n \in \mathbb{N}$, with morphisms $g : Y \rightarrow S$, the projective n -space over S , $\pi : \mathbb{P}_S^n \rightarrow S$ has fibre product $Y \times_S \mathbb{P}_S^n$ satisfies the universal property of $\mathbb{P}_Y^n$.

Indeed, with the definition of $\mathbb{P}_S^n := S \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, we form the following commutative diagram:

$$\begin{array}{ccccc}
 Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Where, both small squares are fibre product diagrams. Also, the morphism $s \circ g : Y \rightarrow \text{Spec}(\mathbb{Z})$ is the unique morphism based on the terminal property of $\text{Spec}(\mathbb{Z})$ in the category of schemes.

Now, given any scheme W together with two morphisms $p_1 : W \rightarrow \mathbb{P}_{\mathbb{Z}}^n$ and $p_2 : W \rightarrow Y$, such that $\varphi \circ p_1 = (s \circ g) \circ p_2$, then the universal property of $\mathbb{P}_S^n$ generates a unique morphism $\psi : W \rightarrow \mathbb{P}_S^n$, such that $p_1 = s' \circ \psi$, and $g \circ p_2 = \pi \circ \psi$, as follow:

$$\begin{array}{c}
 \begin{array}{ccccc}
 W & & & & \\
 \text{---} p_1 \text{---} & & & & \\
 \text{---} \exists! \psi \text{---} & & & & \\
 & \searrow & & & \\
 & & Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 & & \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 & & Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z}) \\
 & & & & & & \\
 & & & & & & \\
 & \swarrow p_2 & & & & & \\
 & & & & & &
 \end{array}
 \end{array}$$

Then, with the universal property of $Y \times_S \mathbb{P}_S^n$, the second equality $g \circ p_2 = \pi \circ \psi$ guarantees a unique morphism $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$, such that $\psi = g' \circ \rho$, and $p_2 = \pi' \circ \rho$, as follow:

$$\begin{array}{c}
 \begin{array}{ccccc}
 W & & & & \\
 \text{---} p_1 \text{---} & & & & \\
 \text{---} \exists! \psi \text{---} & & & & \\
 \text{---} \exists! \rho \text{---} & & & & \\
 & \searrow & & & \\
 & & Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 & & \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 & & Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z}) \\
 & & & & & & \\
 & \swarrow p_2 & & & & & \\
 & & & & & &
 \end{array}
 \end{array}$$

Again, this proves both the existence and the uniqueness of the morphism $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$, such that $p_1 = s' \circ \psi = (s' \circ g') \circ \rho$, and $p_2 = \pi' \circ \rho$. Hence, $Y \times_S \mathbb{P}_S^n$ satisfies the universal property of $\mathbb{P}_Y^n := Y \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, showing $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$.

Now, we'll prove that base change preserves projective morphisms, again by the universality argument: Given a projective morphism $f : X \rightarrow S$, and another morphism $g : Y \rightarrow S$, we know there exists a suitable $n \in \mathbb{N}$, such that there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}_S^n$ and the projection $\pi : \mathbb{P}_S^n \rightarrow S$, such that $f = \pi \circ \iota$.

Which, based on the previous argument, we'll simply denote $Y \times_S \mathbb{P}_S^n$ as $\mathbb{P}_Y^n$. Together with the previous diagrams, we have the following:

$$\begin{array}{ccccc}
 X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n & \xrightarrow{g''} & X & & \\
 \downarrow \iota' & & \downarrow \iota & & \\
 \mathbb{P}_Y^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \downarrow \pi' & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Which, we took the fibre product $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$, and the morphism $\iota' : X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \hookrightarrow \mathbb{P}_Y^n$ is the pullback of the closed immersion $\iota : X \hookrightarrow \mathbb{P}_S^n$, hence ι' is also a closed immersion.

Now, we claim that it satisfies the universal property of $X \times_S Y$: In general, the above proof of preserving projective spaces shows that two consecutive fibre products (consecutive fibre squares in the diagram) generates fibre product for the larger diagram (the large rectangle). By modifying the labels / objects a bit, we indeed get a proof that $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ satisfies the universal property of $X \times_S Y$ as fibre product, hence $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \cong X \times_S Y$.

Finally, by denoting $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ as $X \times_S Y$, the above diagram reduces to the following fibre product diagram:

$$\begin{array}{ccc}
 X \times_S Y & \xrightarrow{g''} & X \\
 \downarrow \iota' & & \downarrow \iota \\
 \mathbb{P}_Y^n & & \mathbb{P}_S^n \\
 \downarrow \pi' & & \downarrow \pi \\
 Y & \xrightarrow{g} & S
 \end{array}
 \quad \begin{array}{c} \curvearrowright f \end{array}$$

Which, the pullback of f is $\pi' \circ \iota' : X \times_S Y \rightarrow Y$, which is precisely a closed immersion into $\mathbb{P}_Y^n$, then a projection onto Y . This shows that the pullback of f is still projective, hence projective morphisms are preserved under base change.

Proof of (b):

Let $\rho : T \rightarrow U$ and $\varphi : U \rightarrow V$ be two projective morphisms, then there exists $n, m \in \mathbb{N}$, such that there exists open immersion $\iota_T : T \hookrightarrow \mathbb{P}_U^n$ and $\iota_U : U \hookrightarrow \mathbb{P}_V^m$, together with projections $\pi_{U,n} : \mathbb{P}_U^n \rightarrow U$ and $\pi_{V,m} : \mathbb{P}_V^m \rightarrow V$, such that the following diagram commutes:

$$\begin{array}{ccccc}
& \mathbb{P}_U^n & & \mathbb{P}_V^m & \\
& \uparrow \iota_T & \searrow \pi_{U,n} & \uparrow \iota_U & \searrow \pi_{V,m} \\
T & \xrightarrow{\rho} & U & \xrightarrow{\varphi} & V
\end{array}$$

Now, let's use some information from part (a): Recall that given any morphism $Y \rightarrow S$ and the projection $\mathbb{P}_S^n \rightarrow S$, the fibre product $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$ (and it's true for arbitrary Y, S and $n \in \mathbb{N}$, as the proof is universal); then, from the above diagram, consider $\varphi : U \rightarrow V$ as the composition $U \xrightarrow{\iota_U} \mathbb{P}_V^m \xrightarrow{\pi_{V,m}} V$, and consider the projection $\pi_{V,n} : \mathbb{P}_V^n \rightarrow V$. Notice by the previous statement, we can construct the projective space $\mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$, as follow:

$$\begin{array}{ccccc}
& \mathbb{P}_{\mathbb{P}_V^m}^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n & \\
& \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} & \\
U & \xrightarrow{\iota_U} & \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V
\end{array}$$

Which, the same argument shows that $\mathbb{P}_U^n \cong U \times_{\mathbb{P}_V^m} \mathbb{P}_{\mathbb{P}_V^m}^n$, as follow:

$$\begin{array}{ccccc}
\mathbb{P}_U^n & \xrightarrow{(\iota_U)'} & \mathbb{P}_{\mathbb{P}_V^m}^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n \\
\downarrow \pi_{U,n} & & \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} \\
U & \xrightarrow{\iota_U} & \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V
\end{array}$$

Which, notice that $(\iota_U)' : \mathbb{P}_U^n \hookrightarrow \mathbb{P}_{\mathbb{P}_V^m}^n$ is a base change of the closed immersion $\iota_U : U \hookrightarrow \mathbb{P}_V^m$, hence is still a closed immersion; composing with the closed immersion $\iota_T : T \hookrightarrow \mathbb{P}_U^n$, we get a closed immersion $(\iota_U)' \circ \iota_T : T \hookrightarrow \mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$, and with the projections $\pi_{V,n} \circ (\pi_{V,m})' : \mathbb{P}_{\mathbb{P}_V^m}^n \rightarrow V$, they compose to be the original morphism $T \rightarrow U \rightarrow V$ (as $(\pi_{V,n} \circ (\pi_{V,m})') \circ ((\iota_U)' \circ \iota_T) = (\pi_{V,n} \circ \pi_{V,m}) \circ (\iota_U \circ \iota_T) = \varphi \circ \rho$ based on the diagrams).

Hence, to prove the composition $\varphi \circ \rho : T \rightarrow V$ is projective, it suffices to show that $\mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$ embeds into some $\mathbb{P}_V^N$, and the projection backs to $\mathbb{V}$ recovers $\varphi \circ \rho$.

Let's first show that there exists a morphism $p : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$. Consider the following diagram:

$$\begin{array}{ccccccc}
& \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n & \longrightarrow & \mathbb{P}_{\mathbb{Z}}^n & \\
(\pi_{V,n})' \downarrow & & & \downarrow \pi_{V,n} & & \downarrow & \\
\mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V & \searrow & & & \\
\downarrow & & & & & \downarrow & \\
\mathbb{P}_{\mathbb{Z}}^m & \xrightarrow{\quad} & \text{Spec}(\mathbb{Z}) & & & &
\end{array}$$

Which, with each „square“ (two are deformed a bit) being fibre squares, the above diagram commutes. Hence, by the universal property of $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ guarantees a unique morphism $P : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$.

Now, consider the Segre Embedding that is suggested in Hartshorne: Given $\mathbb{P}_{\mathbb{Z}}^m = \text{Proj}(\mathbb{Z}[x_0, x_1, \dots, x_m])$ and $\mathbb{P}_{\mathbb{Z}}^n = \text{Proj}(\mathbb{Z}[y_0, y_1, \dots, y_n])$, we cover the fibre product $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ by the open affine subsets $D_+(x_i) \times_{\text{Spec}(\mathbb{Z})} D_+(y_j)$ for $0 \leq i \leq m$ and $0 \leq j \leq n$. Which, with each $D_+(x_i) \cong \text{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, \frac{x_1}{x_i}, \dots, \frac{x_m}{x_i}\right]\right)$ and $D_+(y_j) \cong \text{Spec}\left(\mathbb{Z}\left[\frac{y_0}{y_j}, \frac{y_1}{y_j}, \dots, \frac{y_n}{y_j}\right]\right)$. Hence, their fibre product over $\text{Spec}(\mathbb{Z})$ are given as follow:

$$\begin{aligned}
D_+(x_i) \times_{\text{Spec}(\mathbb{Z})} D_+(y_j) &\cong \text{Spec} \left(\mathbb{Z} \left[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i} \right] \otimes_{\mathbb{Z}} \mathbb{Z} \left[\frac{y_0}{y_j}, \dots, \frac{y_n}{y_j} \right] \right) \\
&\cong \text{Spec} \left(\mathbb{Z} \left[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j} \right] \right)
\end{aligned} \tag{2.1}$$

Then, with $N := nm + n + m$, consider $\mathbb{P}_{\mathbb{Z}}^N = \text{Proj}(\mathbb{Z}[z_0, \dots, z_{ij}, \dots, z_N])$. For any indeterminate z_{ij} , consider its fundamental open subset $D_+(z_{ij}) = \text{Spec}(\mathbb{Z}[\frac{z_0}{z_{ij}}, \dots, \frac{z_N}{z_{ij}}])$, and a ring homomorphism $\psi_{ij} : \mathbb{Z}[\frac{z_0}{z_{ij}}, \dots, \frac{z_N}{z_{ij}}] \rightarrow \mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}]$ as follow:

$$\psi_{ij} \left(\frac{z_{i'j'}}{z_{ij}} \right) = \frac{x_{i'} y_{j'}}{x_i y_j} \tag{2.2}$$

Notice that any $\frac{x_{i'}}{x_i}$ satisfies $\psi_{ij}(\frac{z_{i'j'}}{z_{ij}}) = \frac{x_{i'} y_{j'}}{x_i y_j} = \frac{x_{i'}}{x_i}$, and any $\frac{y_{j'}}{y_j}$ satisfies $\psi_{ij}(\frac{z_{i'j'}}{z_{ij}}) = \frac{x_{i'} y_{j'}}{x_i y_j} = \frac{y_{j'}}{y_j}$. So, it maps onto all the generators of $\mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}]$, showing ψ_{ij} is surjective.

Which, this defines a closed immersion $\psi_{ij}^* : \text{Spec}(\mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}]) = D_+(x_i) \times_{\text{Spec}(\mathbb{Z})} D_+(y_j) \hookrightarrow \text{Spec}(\mathbb{Z}[\frac{z_0}{z_{ij}}, \dots, \frac{z_N}{z_{ij}}]) = D_+(z_{ij})$ (as the ring homomorphism inducing it is surjective). Upon gluing the morphisms using the structure of $\mathbb{P}_{\mathbb{Z}}^N$, this guarantees a closed immersion $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \hookrightarrow \mathbb{P}_{\mathbb{Z}}^N$ (where $N = nm + n + m$).

With the existence of Segre Embedding for projective spaces of $\mathbb{Z}$, say $s : \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \hookrightarrow \mathbb{P}_{\mathbb{Z}}^{nm+n+m}$ (a closed immersion specifically), then with $N := nm + n + m$ consider the following commutative diagram:

$$\begin{array}{ccc}
\mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{p} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\
\downarrow \exists! s' & & \downarrow s \\
\mathbb{P}_V^N & \xrightarrow{\quad} & \mathbb{P}_{\mathbb{Z}}^N \\
\downarrow \pi_{V,N} & & \downarrow \\
V & \xrightarrow{\quad} & \text{Spec}(\mathbb{Z})
\end{array}$$

$\pi_{V,n} \circ (\pi_{V,m})'$ (curved arrow from $\mathbb{P}_V^m \times_V \mathbb{P}_V^n$ to V)

Then, the universality of $\mathbb{P}_V^N = V \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^N$ guarantees a unique morphism $s' : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_V^N$. Finally, we claim that s' is a closed immersion:

Notice in the above diagram, the tall rectangle involving $\mathbb{P}_V^m \times_V \mathbb{P}_V^n, \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n, V, \text{Spec}(\mathbb{Z})$ forms a fibre square by suitable diagram chase.

Then, this in fact enforces $\mathbb{P}_V^m \times_V \mathbb{P}_V^n \cong \mathbb{P}_V^N \times_{\mathbb{P}_{\mathbb{Z}}^N} (\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n)$: Given any scheme W together with two morphisms $w_1 : W \rightarrow \mathbb{P}_V^N$ and $w_2 : W \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, such that the following holds:

$$\begin{array}{ccc}
W & \xrightarrow{w_2} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\
\downarrow \exists! w & & \downarrow s \\
\mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{p} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\
\downarrow \exists! s' & & \downarrow s \\
\mathbb{P}_V^N & \xrightarrow{\quad} & \mathbb{P}_{\mathbb{Z}}^N \\
\downarrow \pi_{V,N} & & \downarrow \\
V & \xrightarrow{\quad} & \text{Spec}(\mathbb{Z})
\end{array}$$

w_1 (curved arrow from W to $\mathbb{P}_V^N$)

Which, the existence and uniqueness of $w : W \rightarrow \mathbb{P}_V^m \times_V \mathbb{P}_V^n$ is guaranteed by the tall fibre rectangle, showing the top square is a fibre square. Since this characterizes $\mathbb{P}_V^m \times_V \mathbb{P}_V^n$ as the fibre product $\mathbb{P}_V^N \times_{\mathbb{P}_Z^N} (\mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n)$, then $s' : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_V^N$ is a base change of $s : \mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n \hookrightarrow \mathbb{P}_Z^N$ (by the morphism $\mathbb{P}_V^N \rightarrow \mathbb{P}_Z^N$), showing s' is a closed immersion (since s is).

Finally, we get this massive commutative diagram:

$$\begin{array}{ccccccc}
 T & \xrightarrow{\iota_T} & \mathbb{P}_U^n & \xrightarrow{(\iota_U)'} & \mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{s'} & \mathbb{P}_V^N \\
 \rho \downarrow & \swarrow \pi_{U,n} & & \swarrow (\pi_{V,n})' & & & \\
 U & \xrightarrow{\iota_U} & \mathbb{P}_V^m & & & & \\
 \varphi \downarrow & \swarrow \pi_{V,m} & & \searrow \pi_{V,N} & & & \\
 V & & & & & &
 \end{array}$$

Which, $\iota_T, (\iota_U)', s'$ are all constructed as closed immersions, hence the composition $T \hookrightarrow \mathbb{P}_V^N$ is also a closed immersion. This realizes $\varphi \circ \rho : T \rightarrow V$ as a projective morphism, hence composition of projective morphisms are still projective.

Finally, back to the original problem, given $f : X \rightarrow S$ and $g : Y \rightarrow S$ two projective morphisms, (a) guarantees their pullbacks to be projective morphisms also. So, $f' : X \times_S Y \rightarrow Y$ and $g' : X \times_S Y \rightarrow X$ are projective morphisms, satisfying the following fibre square:

$$\begin{array}{ccc}
 X \times_S Y & \xrightarrow{f'} & Y \\
 g' \downarrow & & \downarrow g \\
 X & \xrightarrow{f} & S
 \end{array}$$

Then, (b) guarantees the composition of projective morphisms to be projective, hence $f \circ g' = g \circ f' : X \times_S Y \rightarrow S$ is projective, finishing our statement.

3

Problem 3. Let $f : X \rightarrow Y$ be a morphism of separated scheme of finite type over a Noetherian scheme S . Let $Z \subset X$ be a closed subscheme which is proper over S . Show that $f(Z)$ is closed in Y .

Solution. To show $f(Z) \subseteq Y$ is closed, we'll show that with the closed immersion $\iota : Z \hookrightarrow X$, one has $f \circ \iota : Z \rightarrow Y$ is proper, which will show $f \circ \iota(Z) = f(Z)$ is closed in Y .

First, let $\varphi_X : X \rightarrow S$ and $\varphi_Y : Y \rightarrow S$ be the morphism that define X, Y as separated schemes of finite type over S , this implies both φ_X, φ_Y are of finite type and separated morphisms.

Also, consider the closed immersion $\iota : Z \hookrightarrow X$, since Z is proper over S , we have this morphism $\varphi_X \circ \iota : Z \rightarrow S$ be proper.

Now, since $f : X \rightarrow Y$ is a morphism of schemes over S , we have the following commutative diagram:

$$\begin{array}{ccccc}
 & & Z & & \\
 & & \searrow & & \\
 & & \iota & & \\
 & & \searrow & & \\
 & X & \xrightarrow{f} & Y & \\
 & \searrow & & \swarrow & \\
 & \varphi_X & & \varphi_Y & \\
 & & S & &
 \end{array}$$

Or, $\varphi_X = \varphi_Y \circ f$, which implies $\varphi_X \circ \iota = \varphi_Y \circ (f \circ \iota)$. Since $\varphi_X \circ \iota$ is proper, with φ_Y being separated, we have $f \circ \iota$ being proper also (which is a closed map). With Z being closed in itself, $f \circ \iota(Z) = f(Z) \subseteq Y$ is closed. This finishes our proof.

4

Problem 4. Let k be a field and let $R = k[X, Y, Z]/I$, where $I = (X - YZ, XZ - Y^2)$. Let $W = \text{Spec}(R)$.

- Is W a reduced scheme? Justify your answer.
- Is W irreducible? If not, what are its irreducible components?
- Prove or disprove that W has infinitely many rational points if k is infinite (recall that a point $x \in W$ is called rational point if the canonical map $k \rightarrow k(x)$ is an isomorphism, where $k(x)$ is the residue field of the local ring $\mathcal{O}_{W,x}$).

Solution. For the sake of working through all subproblems, we'll first do some calculations regarding two ideals of $k[X, Y, Z]$, defined by $P = (X, Y)$ and $Q = (X - Z^3, Y - Z^2)$. In particular, we claim that they're prime ideals that contains I :

- For $P = (X, Y)$, it's clear that $k[X, Y, Z]/P \cong k[Z]$, hence it's prime; one the other hand, one has $X, Y \in P$ implies $YZ, XZ, Y^2 \in P$, hence $X - YZ, XZ - Y^2 \in P$, showing $I \subseteq P$.
- For Q , we'll show that $k[X, Y, Z]/Q \cong k[T]$ also: Consider the ring homomorphism $\varphi : k[X, Y, Z] \rightarrow k[T]$ by $\varphi(f(X, Y, Z)) = f(T^3, T^2, T)$. Which, this morphism is surjective, as any polynomial $f(T) \in k[T]$ also appears in $k[X, Y, Z]$ (say denotes as $f(Z)$), then $\varphi(f(Z)) = f(T)$. This shows that $k[X, Y, Z]/\ker(\varphi) \cong k[T]$.

Now, we show that $Q = \ker(\varphi)$: For the generators of Q , we have $\varphi(X - Z^3) = T^3 - (T)^3 = 0$, and $\varphi(Y - Z^2) = T^2 - (T)^2 = 0$, hence $Q \subseteq \ker(\varphi)$ (as its generators are contained in their kernel).

For the converse, we claim that all polynomials can be expressed in the form $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ for some $f, g \in k[X, Y, Z]$, and $h(Z) \in k[Z]$: Given any monomial $X^n Y^m Z^p$, we can rewrite it as the following:

$$\begin{aligned} X^n Y^m Z^p &= ((X - Z^3) + Z^3)^n ((Y - Z^2) + Z^2)^m Z^p \\ &= \left(\sum_{k=0}^n \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k} \right) \left(\sum_{l=0}^m \binom{m}{l} (Y - Z^2)^l (Z^2)^{m-l} \right) Z^p \\ &= \left(Z^{3n} + \sum_{k=1}^n \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k} \right) \left(Z^{2m} + \sum_{l=1}^m (Y - Z^2)^l (Z^2)^{m-l} \right) Z^p \end{aligned} \quad (4.1)$$

Which, define $f := \sum_{k=1}^n \binom{n}{k} (X - Z^3)^{k-1} (Z^3)^{n-k}$, and $g := \sum_{l=1}^m (Y - Z^2)^{l-1} (Z^2)^{m-l}$, the above can be rewritten as follow:

$$\begin{aligned} X^n Y^m Z^p &= (Z^{3n} + f \cdot (X - Z^3))(Z^{2m} + g \cdot (Y - Z^2))Z^p \\ &= (f(Z^{2m} + g(Y - Z^2))Z^p) \cdot (X - Z^3) + (gZ^{3n+p}) \cdot (Y - Z^2) + Z^{3n+2m+p} \end{aligned} \quad (4.2)$$

This shows that each monomial can be written as some $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + Z^q$. Hence, with all polynomials being finite k -linear combinations of the monomials, they can all be written in the form of $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ for $f, g \in k[X, Y, Z]$ and $h(Z) \in k[Z]$.

As a consequence, for any polynomial $\ell(X, Y, Z) \in \ker(\varphi)$, since there exists $f, g \in k[X, Y, Z]$ and $h(Z) \in k[Z]$, such that $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$, then the evaluation by φ gives the following:

$$\varphi(\ell(X, Y, Z)) = \varphi(f \cdot (X - Z^3)) + \varphi(g \cdot (Y - Z^2)) + \varphi(h(Z)) = h(T) = 0 \quad (4.3)$$

(Note: recall that $X - Z^3, Y - Z^2 \in \ker(\varphi)$ are verified). Hence, one must have $h(T) = 0$, or $h(Z) = 0$, showing that $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) \in Q$. This proves that $\ker(\varphi) \subseteq Q$, finishing the proof of $\ker(\varphi) = Q$.

Then, since $k[X, Y, Z]/Q \cong k[T]$ based on the homomorphism φ , then Q is a prime ideal.

Finally, to say $I \subseteq Q$, use the algorithm proposed above, some calculation shows the following:

$$\begin{aligned}
 \bullet \quad X - YZ &= X - ((Y - Z^2) + Z^2)Z = (X - Z^3) + Z(Y - Z^2) \in Q & (4.4) \\
 \bullet \quad XZ - Y^2 &= XZ - ((Y - Z^2) + Z^2)^2 \\
 &= XZ - ((Y - Z^2)^2 + 2Z^2(Y - Z^2) + Z^4) \\
 &= XZ - Z^4 - ((Y - Z^2) + 2Z^2)(Y - Z^2) \\
 &= Z(X - Z^3) + (Y + Z^2)(Y - Z^2) \in Q & (4.5)
 \end{aligned}$$

Hence, the generators of I are in Q , showing $I \subseteq Q$.

Now, since $W = \text{Spec}(R) \cong V(I)$ as closed subschemes of $\text{Spec}(k[X, Y, Z])$, we'll work in this regime.
Here are the solutions to the subproblems:

- (a) Here, we claim that W is reduced, by proving that R is reduced. In particular, we'll show that $I = P \cap Q$ to show that it's a radical.

It's clear that $I \subseteq P \cap Q$ based on the previous calculation. To show that $P \cap Q \subseteq I$, we'll consider the algorithm used when proving $k[X, Y, Z]/Q \cong k[T]$.

First, for any polynomial $s \in P \cap Q$, since $s \in P = (X, Y)$, so it can be written in the form $s = v_1X + v_2Y$ for some $v_1, v_2 \in k[X, Y, Z]$.

Then, perform the algorithm for Q on v_1, v_2 , there exists $f_1, f_2, g_1, g_2 \in k[X, Y, Z]$, and $h_1(Z), h_2(Z) \in k[Z]$, such that the following holds:

$$\begin{aligned}
 s &= v_1X + v_2Y \\
 &= (f_1(X - Z^3) + g_1(Y - Z^2) + h_1(Z))X + (f_2(X - Z^3) + g_2(Y - Z^2) + h_2(Z))Y & (4.6) \\
 &= f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y
 \end{aligned}$$

Which, the first four terms in the sum is in PQ (since they're multiples of products of the generators of $P = (X, Y), Q = (X - Z^3, Y - Z^2)$). So, in case for $s \in P \cap Q$, one needs $h_1(Z)X + h_2(Z)Y \in P \cap Q$.

It's true that $h_1(Z)X + h_2(Z)Y \in P$, hence the only condition is $h_1(Z)X + h_2(Z)Y \in Q$. Which, we can rewrite it as follow:

$$\begin{aligned}
 h_1(Z)X + h_2(Z)Y &= h_1(Z)((X - Z^3) + Z^3) + h_2(Z)((Y - Z^2) + Z^2) \\
 &= h_1(Z)(X - Z^3) + h_2(Z)(Y - Z^2) + (h_1(Z)Z^3 + h_2(Z)Z^2) & (4.7)
 \end{aligned}$$

Which, one requires $h_1(Z)Z^3 + h_2(Z)Z^2 \in Q = (X - Z^3, Y - Z^2)$. Then, this imposes that $h_1(Z)Z^3 + h_2(Z)Z^2 = 0$ (as it's independent from X, Y). So, one has $Z^2(h_1(Z)Z + h_2(Z)) = 0$, showing $h_2(Z) = -h_1(Z)Z$.

As a result, the term $h_1(Z)X + h_2(Z)Y$ becomes the following:

$$h_1(Z)X - h_1(Z)YZ = h_1(Z)(X - YZ) \in I \quad (4.8)$$

So, this shows that the last two terms in $s = f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y$ is contained in I . To show $s \in I$, one simply needs the first four terms to be in I also.

Since $PQ = (X, Y)(X - Z^3, Y - Z^2) = ((X - Z^3)X, (Y - Z^2)X, (X - Z^3)Y, (Y - Z^2)Y)$, it suffices to prove these generators are in I :

$$\begin{aligned} \bullet \quad (X - Z^3)X &= X^2 - (XZ)Z^2 = X^2 - (XZ)Z^2 + -Y^2Z^2 + Y^2Z^2 \\ &= (X^2 - (YZ)^2) - Z^2(XZ - Y^2) \\ &= (X + YZ)(X - YZ) = Z^2(XZ - Y^2) \in I \end{aligned} \quad (4.9)$$

$$\begin{aligned} \bullet \quad (Y - Z^2)X &= XY - (XZ)Z = XY - (XZ)Z + Y^2Z - Y^2Z \\ &= Y(X - YZ) - Z(XZ - Y^2) \in I \end{aligned} \quad (4.10)$$

$$\begin{aligned} \bullet \quad (X - Z^3)Y &= XY - YZ^3 + XZ^2 - XZ^2 \\ &= Z^2(X - YZ) + XY - XZ^2 - Y^2Z + Y^2Z \\ &= Z^2(X - YZ) + Y(X - YZ) - Z(XZ - Y^2) \\ &= (Z^2 + Y)(X - YZ) - Z(XZ - Y^2) \in I \end{aligned} \quad (4.11)$$

$$\begin{aligned} \bullet \quad (Y - Z^2)Y &= Y^2 - YZ^2 - XZ + XZ \\ &= Z(X - YZ) - (XZ - Y^2) \in I \end{aligned} \quad (4.12)$$

This shows the generators of PQ are in I , hence $PQ \subseteq I$. As a consequence, the first four terms of $s = f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y$ is also in I .

This proves $s \in I$, showing $P \cap Q \subseteq I$. Hence, $I = P \cap Q$, showing it's a radical. This shows that $R = k[X, Y, Z]/I$ is reduced, hence $W = \text{Spec}(R)$ is also reduced.

- (b) We show that W is reducible, in particular that $W = V(I) = V(P) \cup V(Q)$, where $V(P), V(Q)$ are its irreducible components.

First, since $I \subseteq P$ and $I \subseteq Q$, it's clear that $V(P), V(Q) \subseteq V(I)$, hence $V(P) \cup V(Q) \subseteq V(I)$; to claim the reverse inclusion, we'll consider the following element:

$$Z(X - YZ) - (XZ - Y^2) = XZ - YZ^2 - XZ + Y^2 = Y(Y - Z^2) \in I \quad (4.13)$$

Suppose $K \in V(I)$ is a prime ideal containing I , then $Y(Y - Z^2) \in K$, which the prime property guarantees $Y \in K$ or $(Y - Z^2) \in K$.

- If $Y \in K$, then since $X - YZ \in I \subseteq K$, one has $X = (X - YZ) + YZ \in K$, showing $P = (X, Y) \subseteq K$, or $K \in V(P)$.
- Else if $Y - Z^2 \in K$, then since $X - YZ \in I \subseteq K$ again, one has $X - Z^3 = (X - YZ) + (YZ - Z^3) = (X - YZ) + Z(Y - Z^2) \in K$, showing $Q = (X - Z^3, Y - Z^2) \subseteq K$, or $K \in V(Q)$.

Hence, in either case $K \in V(P) \cup V(Q)$, showing $W = V(I) = V(P) \cup V(Q)$.

To show that W is reducible, we'll show that $V(P), V(Q)$ are proper closed subsets of W : Suppose the contrary that one of them is not proper closed subset of W , there are two cases:

- If $V(P) = W$, this shows that $V(Q) \subseteq V(P)$; with P, Q being prime (which both are radicals), this shows that $Q \supseteq P$. However, if consider the ring homomorphism $\psi: k[X, Y, Z] \twoheadrightarrow k$ by $\psi(X, Y, Z) = \psi(1, 1, 1)$, notice that $\psi(X - Z^3) = 1 - 1^3 = 0$ and $\psi(Y - Z^2) = 1 - 1^2 = 0$, showing $Q = (X - Z^3, Y - Z^2) \subseteq \ker(\psi)$; yet, if plugin the generators of P , we get $\psi(X) = \psi(Y) = 1 \neq 0$, showing $P \not\subseteq \ker(\psi)$, which contradicts our construction that $\ker(\varphi) \supseteq Q \supseteq P$.

- Else if $V(Q) = W$, this shows that $V(P) \subseteq V(Q)$, which implies $P \supseteq Q$. However, consider the ring homomorphism $\rho : k[X, Y, Z] \twoheadrightarrow k$ by $\rho(f(X, Y, Z)) = f(0, 0, 1)$, notice that $\rho(X) = \rho(Y) = 0$, showing $P = (X, Y) \subseteq \ker(\rho)$; yet, if plugin the generators of Q , we get $\rho(X - Z^3) = 0 - 1^3 = -1 \neq 0$, showing $Q \not\subseteq \ker(\rho)$, again this contradicts our construction that $\ker(\rho) \supseteq P \supseteq Q$.

Since in either case we get a contradiction, then $V(P), V(Q)$ cannot be the whole W , showing they're proper closed subsets, which proves that $W = V(P) \cup V(Q)$ is reducible.

Finally, to show $V(P), V(Q)$ are its irreducible components, consider any other irreducible subsets $W' \subseteq W = V(I)$, since W' is also closed in $\text{Spec}(k[X, Y, Z])$ (by the closeness of $V(I)$ in it), then $W' = V(K)$ for some ideal $K \subseteq k[X, Y, Z]$. Which, the irreducibility of W' guarantees K to be a prime ideal. However, if $V(K) \subseteq V(I)$, one has $I \subseteq \sqrt{I} \subseteq K$ (by primeness, it is a radical), showing that $K \in V(I) = V(P) \cup V(Q)$. This implies that either $K \in V(P)$ (implying $P \subseteq K$, or $V(P) \supseteq V(K)$), or $K \in V(Q)$ (implying $Q \subseteq K$, or $V(Q) \supseteq V(K)$), showing that $W' = V(K)$ is contained in $V(P)$ or $V(Q)$. Hence, these two must be the irreducible components of W .

- (c) With k being infinite, we'll show that there are infinitely rational points in W . For any $t \in k$, consider the maximal ideal $\mathfrak{m}_t := (X, Y, Z - t) \subset k[X, Y, Z]$. Notice that it contains I , since $X - YZ, XZ - Y^2 \in \mathfrak{m}_t$ by the fact that $X, Y \in \mathfrak{m}_t$. Hence, $x := \mathfrak{m}_t$ can be realized as a maximal ideal in $R = k[X, Y, Z]/I$ by descending to the quotient, which is realized as a closed point in $W = \text{Spec}(R)$.

To compute its residue field, let $\overline{\mathfrak{m}}_t \subseteq R$ denotes the quotient of $\mathfrak{m}_t$. Then, its corresponding residue field is as follow:

$$k(\overline{\mathfrak{m}}_t) = \frac{R_{\overline{\mathfrak{m}}_t}}{\overline{\mathfrak{m}}_t R_{\overline{\mathfrak{m}}_t}} \cong \left(\frac{R}{\overline{\mathfrak{m}}_t} \right)_{\overline{\mathfrak{m}}_t} \cong k \quad (4.14)$$

Note that since $I \subseteq \mathfrak{m}_t$, then the composition of projections $\varphi : k[X, Y, Z] \twoheadrightarrow R \twoheadrightarrow R/\overline{\mathfrak{m}}_t$ has the kernel being precisely $\varphi^{-1}(\overline{\mathfrak{m}}_t) = \mathfrak{m}_t$, hence $k \cong k[X, Y, Z]/\mathfrak{m}_t \cong R/\overline{\mathfrak{m}}_t$ (where, the first isomorphism can be realized by the evaluation $e : k[X, Y, Z] \twoheadrightarrow k$ by $e(f(X, Y, Z)) = f(0, 0, t)$, where the kernel is precisely $\mathfrak{m}_t = (X, Y, Z - t)$).

Hence, with k being infinite, there are infinitely many $t \in k$, with $\overline{\mathfrak{m}}_t \subseteq R$ being a maximal ideal (or a closed point in $W = \text{Spec}(R)$), such that the residue field is isomorphic to k . Hence, W has infinitely many rational points.

5

Problem 5. Let $f : X \rightarrow Y$ be a surjective morphism between two schemes which are finite type and separated integral schemes over a field. Assume that X is an affine scheme. Prove or disprove that Y is an affine scheme if and only if it is a quasi-affine scheme (recall that quasi-affine means open subscheme of an affine scheme).

Solution. We'll prove that Y is affine $\iff Y$ is quasi-affine.

$\implies$:

Suppose Y is affine, then it's an open subscheme of Y (an affine scheme), hence Y is quasi-affine.

$\impliedby$:

Suppose that Y is quasi-affine. Here, we'll denote the global section of Y as $A := \mathcal{O}_Y(Y)$, and let B be a commutative ring satisfies $X \cong \text{Spec}(B)$ (which, X, Y being integral schemes implies A, B are integral domains).

Let's first deal with a special case: Suppose B is a field, then $X = \text{Spec}(B)$ is a one point scheme, hence with $f : X \rightarrow Y$ being surjective, Y must be a one point scheme also. Take any affine neighborhood of the only point of Y , it must be Y itself, showing that $Y = \text{Spec}(A')$ for some ring A' (in particular, $A' = \mathcal{O}_Y(Y) = A$). So, Y is affine.

From now on, we'll assume B (the global section of X) is not a field.

Let k be a field where X, Y are finite type and separated integral schemes over it. Which, given $\varphi_X : X \rightarrow \text{Spec}(k)$, $\varphi_Y : Y \rightarrow \text{Spec}(k)$ as the corresponding morphisms, φ_X, φ_Y are finite type and separated morphisms that satisfy the following diagram:

$$\begin{array}{ccc} X = \text{Spec}(B) & \xrightarrow{f} & Y \\ & \searrow \varphi_X & \swarrow \varphi_Y \\ & \text{Spec}(k) & \end{array}$$

Then, on the global section level, this reverses to the following diagram of ring homomorphisms:

$$\begin{array}{ccc} B & \xleftarrow{f^\#} & \mathcal{O}_Y(Y) = A \\ & \swarrow \varphi_X^\# & \searrow \varphi_Y^\# \\ & k & \end{array}$$

Now, we claim the following statements in order:

- (a) Y can be realized as open subschemes of $\text{Spec}(A)$.
- (b) Y is affine, in particular $Y \cong \text{Spec}(A)$.

Proof of (a):

Since Y is assumed to be quasi-affine, there exists an open immersion $\iota_Y : Y \hookrightarrow \text{Spec}(R)$ for some commutative ring R . Then, since the affinization of Y , say $\text{Aff}_Y = \text{Spec}(A)$, with the natural morphism of schemes $i_Y : Y \rightarrow \text{Aff}_Y$, there exists a unique morphism of schemes $\psi_Y : \text{Aff}_Y \rightarrow \text{Spec}(R)$, such that the following commutative diagram holds:

$$\begin{array}{ccc}
Y & \xrightarrow{i_Y} & \text{Aff}_Y \\
\searrow \iota_Y & & \swarrow \exists! \psi_Y \\
& \text{Spec}(R) &
\end{array}$$

Which, since ι_Y is injective (as set map), this enforces i_Y to also be injective also.

Now, consider the ring homomorphism $\psi_Y^\# : R \rightarrow A$. For any $P \in Y \hookrightarrow \text{Spec}(R)$, there exists some element $f \in R$, such that $P \in D(f) \subseteq Y$ (since Y is open subscheme of $\text{Spec}(R)$). Then, denote $f' := \varphi_Y^\#(f) \in A$. Using the Hartshorne Notation, we can consider the subset $Y_{f'} \subseteq Y$ (denoting all points $y \in Y$, such that the localization $f'_y \in \mathcal{O}_{Y,y}$ is invertible, or not in the maximal ideal $\mathfrak{m}_y \subseteq \mathcal{O}_{Y,y}$). Notice that this set is precisely $Y_{f'} = D(f)$, since when passing to the stalk, any $P \in D(f)$ has $\mathcal{O}_{Y,P}$ has f being an invertible element, hence with f' being the image of f in A , under localization it's also invertible.

As a result, one has $R_f \cong \mathcal{O}_{\text{Spec}(R)}(D(f)) = \mathcal{O}_Y(D(f)) = \mathcal{O}_Y(Y_{f'}) = \mathcal{O}_Y(D(f')) \cong A_{f'}$, showing that $P \in D(f) \subseteq Y$ has an affine neighborhood $D(f) \cong D(f') \cong \text{Spec}(A_{f'})$ in Y .

This shows that $i_Y : Y \hookrightarrow \text{Spec}(A)$ is actually an open immersion (as any point in Y has an open neighborhood in Y , such that it's scheme-isomorphic to a fundamental open subset of $\text{Spec}(A)$). So, Y is an open subscheme of $\text{Spec}(A)$.

Proof of (b):

Now, together with the open immersion $i_Y : Y \hookrightarrow \text{Spec}(A)$, the morphism $\varphi_Y : Y \rightarrow \text{Spec}(k)$ uniquely factors through some morphism $\rho : \text{Spec}(A) \rightarrow \text{Spec}(k)$ (since $\text{Spec}(A) = \text{Aff}_Y$ the affinization), which we have the following commutative diagram of morphisms:

$$\begin{array}{ccccc}
X & \xrightarrow{f} & Y & \xrightarrow{i_Y} & \text{Spec}(A) \\
\searrow \varphi_X & & \downarrow \varphi_Y & & \swarrow \exists! \rho \\
& & \text{Spec}(k) & &
\end{array}$$

Where the morphism $i_Y \circ f$ is separated (as it's morphism between affine schemes). Which, on the level of global section, we have the following commutative diagram:

$$\begin{array}{ccccc}
B & \xleftarrow{f^\#} & A & \xleftarrow{\text{id}_A} & A \\
\swarrow \varphi_X^\# & & \uparrow \varphi_Y^\# & & \searrow \rho^\# \\
& & k & &
\end{array}$$

Where the middle A is given by the global section of Y , $A = \mathcal{O}_Y(Y)$.

Then, this implies that the morphism $i_Y \circ f : X = \text{Spec}(B) \rightarrow \text{Spec}(A)$ is induced by the ring homomorphism $f^\# \circ \text{id}_A = f^\# : A \rightarrow B$; and, for the diagram to commute, $f^\#$ is in fact a k -algebra homomorphism. Moreover, $f^\#$ is of finite type, as B is a finitely generated k -algebra, and the morphism is a k -algebra homomorphism. This realizes B as a finitely generated A -algebra also.

Now, notice that $i_Y \circ f$ is a dominant morphism: Because A is an integral domain, then $\text{Spec}(A)$ is irreducible, hence $Y \subseteq \text{Spec}(A)$ as an open subscheme is dense and irreducible (or $\overline{Y} = \text{Spec}(A)$). With f being surjective, one has $f(X) = Y$, hence $\overline{i_Y \circ f(X)} = \overline{i_Y(Y)} = \overline{Y} = \text{Spec}(A)$. Hence, with the morphism $i_Y \circ f$ being dominant, the induced ring homomorphism $f^\#$ is injective.

Finally, we claim that $Y = \text{Spec}(A)$: Since $Y \subseteq \text{Spec}(A)$ is open, there exists $a_i \in A$ (where $i \in I$ for some index set), such that $Y = \bigcup_{i \in I} D(a_i)$. Since $\varphi_Y : Y \rightarrow \text{Spec}(k)$ is of finite type (which is quasi-compact), with $\text{Spec}(k)$ being quasi-compact, this implies $Y = \varphi_Y^{-1}(\text{Spec}(k))$ is quasi-compact. Hence, one can choose the cover to be finite, say $a_1, \dots, a_n \in A$ satisfies $Y = \bigcup_{i=1}^n D(a_i)$.

Then, with $f : X \twoheadrightarrow Y$ and the induced ring homomorphism $f^\# : A \hookrightarrow B$, one has $\text{Spec}(B) = X = f^{-1}(Y) = \bigcup_{i=1}^n f^{-1}(D(a_i)) = \bigcup_{i=1}^n D(f^\#(a_i))$. So, since $D(f^\#(a_i))$ forms an open cover of $\text{Spec}(B)$, this implies $B = (f^\#(a_1), \dots, f^\#(a_n))$.

At this point I was aiming for $A = (a_1, \dots, a_n)$ (currently in progress), then with $\bigcup_{i=1}^n D(a_i) = \text{Spec}(A)$ based on this argument, it'll finish the proof that $Y = \text{Spec}(A)$, or showing Y is affine.